Contact

anna@astroram.ru

www.linkedin.com/in/astroram (LinkedIn)

Top Skills

Python (Programming Language)

Analytical Skills

IDL

Languages

Английский (Professional Working)

Испанский (Elementary)

Honors-Awards

Presidential Grant of the Russian Federation for Young Scientists

RFBR Grant for Graduate Students — Atmospheric Parameters of Magnetic Chemically Peculiar Stars

Russian Science Foundation Grant — Silicon in the Atmospheres of Ap Stars

Publications

Slowly Rotating Peculiar Star BD +00°1659 as a Benchmark for Stratification Studies in Ap/Bp Stars

Distinct Barium Isotope Ratios in CEMP-s and CEMP-rs Stars

Evolutionary Status of the Ap Stars HD 110066 and HD 153882

Fundamental Parameters of the Ap-Stars GO And, 84 UMa, and κ Psc

Non-LTE Abundance Analysis of A–B Stars with Low Rotational Velocities. II. Do A–B Stars with Normal Abundances Exist?

Anna Romanovskaya

Scientific Researcher | PhD in astrophysics

Moscow, Moscow City, Russia

Summary

Researcher specializing in high-resolution stellar spectroscopy and stellar atmosphere modelling. Experienced in determining atmospheric parameters and chemical abundances in magnetic chemically peculiar stars, metallic-lined stars, normal stars, and spectroscopic binary systems.

My work includes spectral modelling, analysis of high-resolution spectroscopic observations, and interpretation of stellar spectra. My research interests also include hybrid approaches that combine classical spectral synthesis with machine learning and numerical optimization for the analysis of binary star spectra.

My publications are below:
https://scholar.google.com/citations?user=BJx14fIAAAAJ&hl=ru&oi=sra

Stack:
Python | Scientific computing | Data analysis pipelines | Numerical optimization | Machine learning methods | LLmodels | SynthV / Synmast | VALD3 database

Experience

Institute of Astronomy Russian Acad. Sci.

11 years 6 months

Scientific Researcher

January 2025 - Present (1 year 6 months)

Moscow, Russia

Research in high-resolution stellar spectroscopy focused on the analysis of chemically peculiar stars, Am stars, normal stars, and spectroscopic binary systems.

- Analysis of high-resolution stellar spectra and determination of atmospheric parameters and chemical abundances
- Spectroscopic modelling of magnetic and non-magnetic stars
- Development and application of optimization and machine learning methods for stellar spectral analysis
- Processing and analysis of large spectroscopic datasets using Python

Preparation of scientific publications, participation at conferences

Junior science researcher

January 2017 - January 2025 (8 years 1 month)

Russia, Moscow

Research on chemically peculiar stars using spectroscopic observations and stellar atmosphere modelling.

- Determination of atmospheric parameters and chemical abundances in chemically peculiar stars
- Spectral modelling including LTE and non-LTE effects
- Participation in preparation of publications
- Presentations at national and international scientific conferences

PhD (2022): Determination of Fundamental Parameters of Magnetic Chemically Peculiar Stars Using Spectroscopic Methods

Research Assistant

January 2015 - January 2017 (2 years 1 month)

Moscow, Russia

Early research experience in stellar spectroscopy and analysis of chemically peculiar stars.

- Determination of atmospheric parameters of magnetic chemically peculiar stars using photometric indices
- Chemical abundance analysis of rare-earth elements
- Spectroscopic analysis of two chemically peculiar stars for diploma research (fundamental parameters and chemical abundances)
- Participation in preparation of scientific publications

Education

Yandex School of Data Analysis

Artificial Intelligence in Natural Sciences · (October 2025 - October 2027)

Institute of astronomy RAS

PhD, Astronomy and astrophysics · (October 2017 - October 2021)

Moscow State University

Student, Astronomy and Astrophysics · (September 2011 - September 2016)